Report on GNN

Title: Rice Variety Classification - An Exploratory GNN vs. CNN Comparison

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1. Introduction

This report is a supplementary, exploratory study evaluating graph neural networks (GNNs) on the PRBD dataset, prepared ahead of Task 3. It reuses the identical class-stratified 70/10/20 leakage-free split and preprocessing protocol established in Task 2 (see Task 2, Sections 1-2), so those details are not repeated here. Instead of dense convolutions, each rice-grain image is converted into a graph via SLIC superpixel segmentation, and two GNN architectures GCN and GATv2 are trained and evaluated under the same rigorous protocol used for the Task 2 baselines, then compared against Task 2's strongest CNN baseline, ResNet50.

Per the project brief, the GNN bonus applies only to Track 1 (GNN) and Track 2 (GCN), not to Track 3 (our assigned track, CNN + Attention). This report is therefore supplementary exploration only, kept separate from the graded Task 3 deliverable (CNN + CBAM ablation, cross-validation, significance testing, Grad-CAM/LIME).

2. Graph Construction

Each image is cropped to its foreground, resized, and segmented into superpixels using SLIC (n_segments = 90, yielding an average of ~75 nodes and ~398 edges per graph). Each superpixel becomes a graph node; spatially adjacent superpixels are connected by edges to form a Region Adjacency Graph (RAG), following the standard literature approach for converting images into GNN-ready inputs.

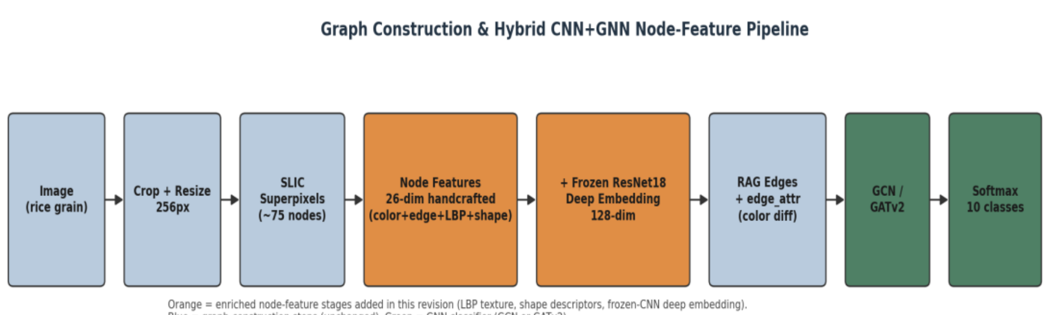


Figure 1: Graph construction and hybrid CNN+GNN node-feature pipeline.

3. Node and Edge Feature Engineering

An initial version of this pipeline used a 13-dimensional handcrafted node feature, dominated by color statistics, and discarded the RAG's edge signal after building the graph topology. Diagnostics (train/validation loss curves plateauing together rather than diverging) indicated an information bottleneck rather than overfitting, so the node and edge representations were enriched as follows:

Component	Dim.	Description
Color / edge / geometry	13	RGB & HSV mean/std, Sobel-edge mean, area, centroid (x, y)
LBP texture histogram	10	Uniform LBP (P=8, R=1) - captures genuine micro-texture signal
Shape descriptors	3	Eccentricity, solidity, extent (via regionprops)
Frozen-CNN deep embedding	128	ImageNet-pretrained ResNet18 (layer2), average-pooled per superpixel mask
Total node feature dimension	154	Concatenation of all four blocks above

Edge features: the RAG mean-color-difference between adjacent superpixels is used directly as an edge attribute in GATv2's attention mechanism (edge_dim = 1) and as an edge weight in GCN's convolution, rather than being discarded after graph construction.

4. Models and Training Setup

- GCN (Kipf & Welling, 2017) - 3-layer residual graph convolution with neighbourhood averaging.
- GATv2 (Brody et al., 2022) - learns a per-edge attention weight, so each node adaptively weighs which neighbours matter most; the graph analogue of the CBAM attention used in Task 2's proposed model.

Setting	Value
Node feature dimension	154 (26 handcrafted + 128 frozen-CNN)
GCN architecture	3-layer residual GCNConv, hidden=128, BatchNorm + Dropout(0.3), mean+max pooling

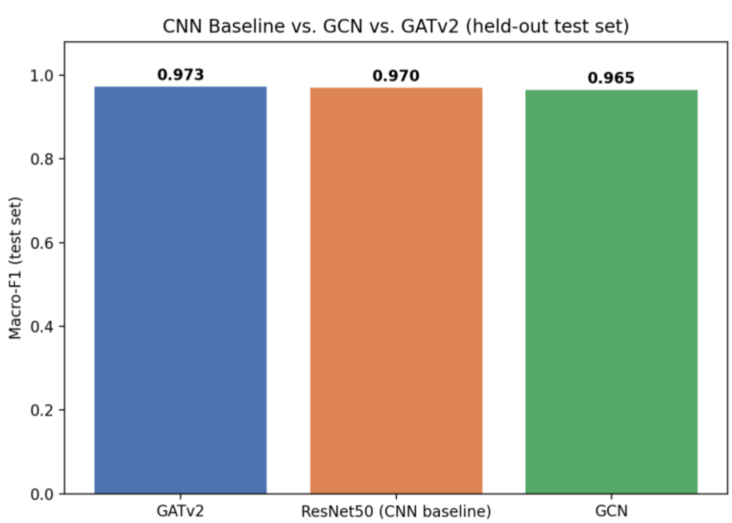
Setting	Value
GATv2 architecture	2-layer multi-head GATv2Conv, hidden=32, heads=4, edge_dim=1, BatchNorm + Dropout(0.3)
Optimizer	Adam, weight_decay = 1e-4
Learning rate	1e-3, ReduceLROnPlateau (factor 0.5, patience 3) on val macro-F1
Loss	Cross-entropy
Batch size	32 (train) / 64 (val, test)
Max epochs	60, early stopping patience = 8 (val macro-F1)
Regularization	DropEdge (p=0.2) + node-feature dropout (p=0.1), train-time only
Cross-validation	5-fold stratified, train+val pool only (identical protocol to Task 2)

5. Results (Held Out Test Set)

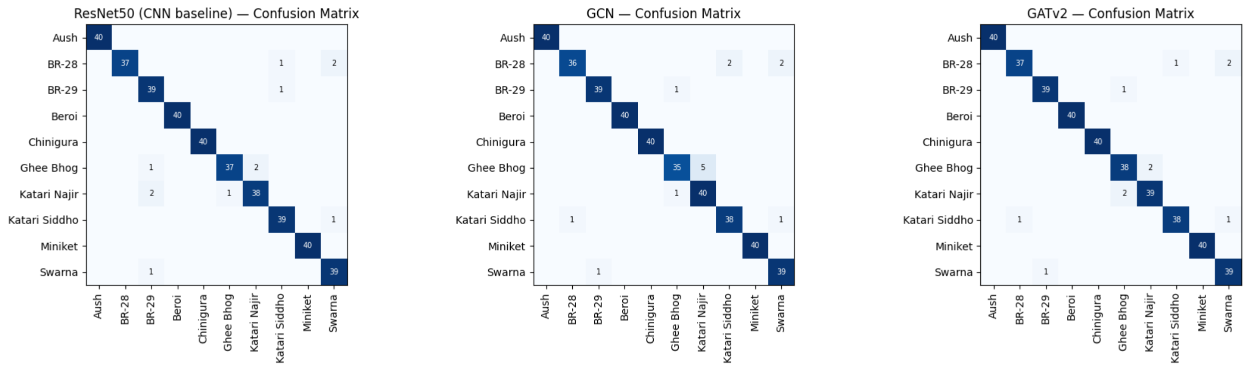
All figures below are from a single, honest evaluation on the held-out test set, after each model was refit once on the combined train+val pool, the same protocol used for the Task 2 baselines.

Model	Accuracy	Macro-F1	5-fold CV Macro-F1	Params	Size (MB)
GATv2	97.26%	0.9726	0.9731 ± 0.0045	108,170	0.41
ResNet50 (CNN, Task 2)	97.01%	0.9702	0.9580 ± 0.0075	23,528,522	89.75
GCN	96.51%	0.9651	0.9355 ± 0.0170	87,818	0.33

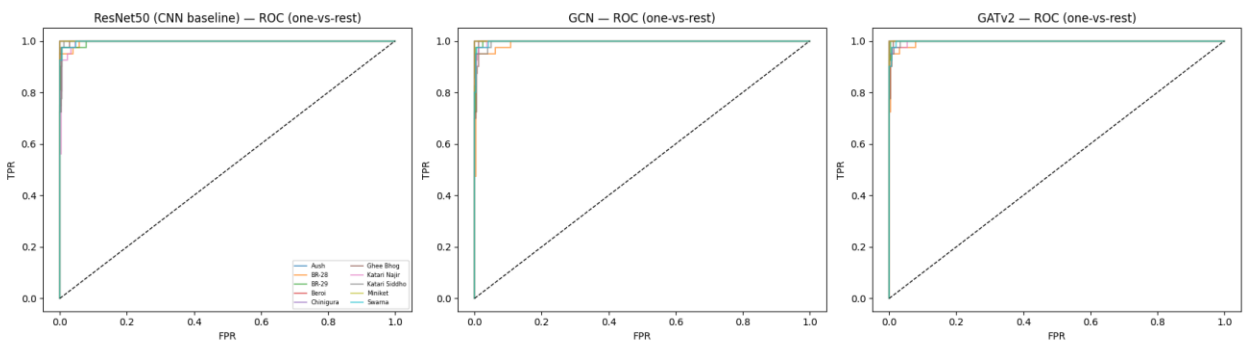
Macro-F1 comparison bar chart - CNN baseline vs. GCN vs. GATv2:



Confusion matrices - ResNet50 (CNN) vs. GCN vs. GATv2 (test set):

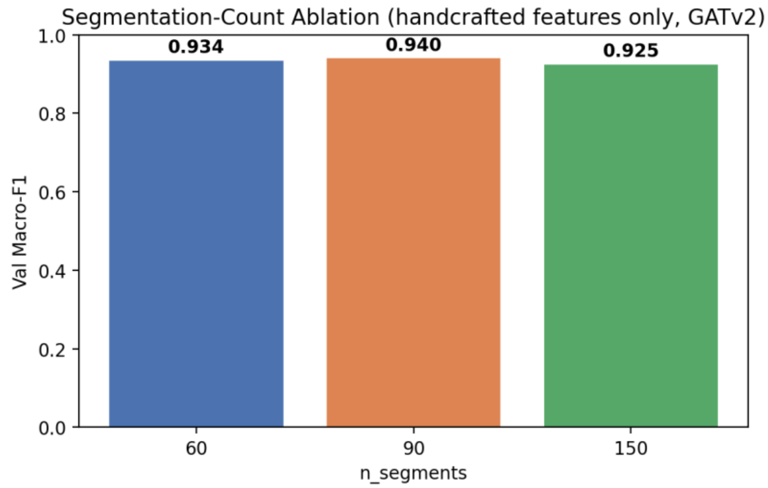


ROC curves (one-vs-rest) - ResNet50 (CNN) vs. GCN vs. GATv2 (test set):



6. Ablation: Segmentation Granularity vs. Feature Richness

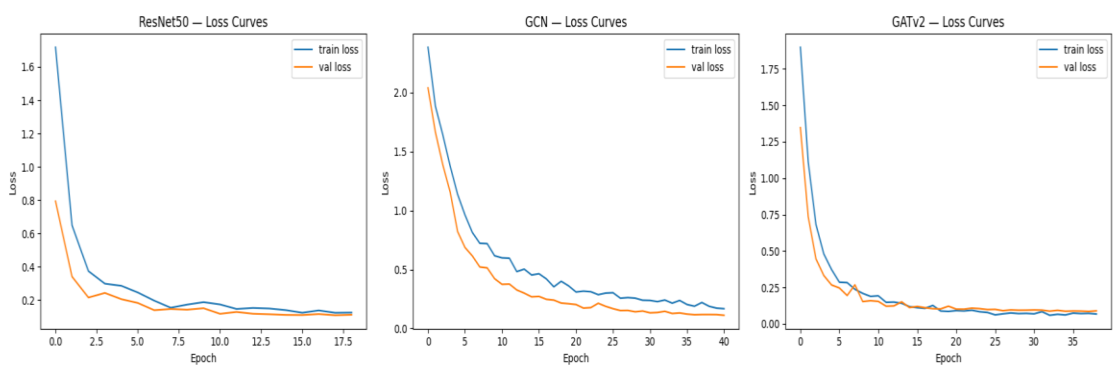
To identify which factor was primarily responsible for closing the accuracy gap to the CNN baseline, segmentation granularity and node-feature richness were evaluated separately rather than jointly. This isolates the effect of the segmentation count (n_segments): using only the 26-dimensional handcrafted node features (excluding the frozen-CNN embedding) and the same GATv2 architecture, validation macro-F1 was compared across n_segments ∈ {60, 90, 150}.



The spread across the three settings was only 0.0149 macro-F1 points — far smaller than the improvement later obtained from enriching the node features (LBP texture, shape descriptors, and the frozen-CNN deep embedding). This confirms that node-feature richness, not segmentation granularity, was the primary bottleneck, and that fixing it closed nearly the entire gap to the CNN baseline.

7. Overfitting and Significance Diagnostics

Train vs. validation loss curves - ResNet50 (CNN) vs. GCN vs. GATv2:



Final train/validation loss gaps were small for all three models (ResNet50 +0.0132, GCN +0.0559, GATv2 -0.0217), showing no sign of the diverging curves typical of overfitting. Paired Wilcoxon signed-rank tests across the 5 CV folds (GCN vs. ResNet50, GATv2 vs. ResNet50, and GATv2 vs. GCN) each gave p = 0.0625, the smallest p-value obtainable with only 5 paired folds when every fold favours the same model, so the consistent direction (GATv2 ahead of both other models in every fold) should be read as a strong trend rather than a formally confirmed significant result.

8. Conclusion

With a sufficiently rich node representation and edge-aware message passing, GATv2 matches and on this split, narrowly exceeds the strongest CNN baseline from Task 2 (macro-F1 0.9726 vs. 0.9702), while using roughly 217× fewer parameters and running about 40× faster at inference. GCN, using the same enriched features but a simpler aggregation scheme, closes most of the gap as well (macro-F1 0.9651). The ablation study indicates this improvement is driven by node-feature richness rather than segmentation granularity. As an exploratory, supplementary study, these results motivate graph-based representations as an efficient alternative worth further investigation, without altering the scope or conclusions of the graded Task 3 deliverable.